

Xin (Allen) Wang

allen.wang.xw532@yale.edu | LinkedIn | Website

EDUCATION

Yale University

Ph.D. Computational Biology and Bioinformatics

Aug 2025 – May 2030

New Haven, CT

Vanderbilt University

M.S. Computer Science, GPA: 4.0/4.0

Aug 2023 – May 2025

Nashville, TN

- Thesis Track: *Data-driven Machine Learning for Small Molecule Drug Discovery*

University of California, Santa Barbara

B.S. Statistics and Data Science, Overall GPA: 3.88/4.0, Major GPA: 3.92/4.0

Aug 2019 – Jun 2023

Goleta, CA

RESEARCH EXPERIENCE

Biomedical Informatics & Data Science

Ph.D. Student

Advisor: Dr. Qingyu Chen

Research Interests: AI Systems for Scientific Discovery

Sep 2025 – Present

Yale University

Center for AI in Protein Dynamics

Researcher

Advisor: Dr. Carlos Oliver

Research Interests: Protein & RNA Language Models, Biomolecular Design

May 2025 – Present

Vanderbilt University

Network and Data Science (NDS) Lab

Graduate Research Assistant

Advisors: Dr. Tyler Derr, Dr. Jens Meiler

Research Interests: AI for Science, Data-centric AI

Aug 2023 – May 2025

Vanderbilt University

Caves Lab/Data Science Capstone

Undergraduate Research Assistant

Advisor: Dr. Eleanor Caves

Research Interests: Computer Vision, Biological Image Analysis

Nov 2022 – Jun 2023

UC Santa Barbara

PUBLICATIONS

[†]Equal contribution

1. **Xin Wang**, Kaiwen Shi, and Carlos Oliver. “BioBlobs: Unsupervised Discovery of Functional Substructures for Protein Function Prediction.” In *Advances in Neural Information Processing Systems (NeurIPS ’26)*.
2. **Xin Wang**, Yu Wang, Yunchao Liu, Jens Meiler, and Tyler Derr. “Scaffold-Aware Generative Augmentation and Reranking for Enhanced Virtual Screening.” In *Proceedings of the SIAM International Conference on Data Mining (SDM ’26)*.
3. Tianyu Liu[†], **Xin Wang**[†], Antonia Panescu[†], et al. “Benchmarking AI Agents for Addressing Scientific Challenges Across Scales.” Submitted to *Nature*, 2026. arXiv:2606.12736.
4. Qingyu Chen, **Xin Wang**, Rong Zhou, et al. “AI Scientists in Health: Trustworthy Agentic AI for Scientific Inquiry.” Submitted to *Nature Biomedical Engineering*, 2026. SSRN:6889798.
5. Yunchao Liu[†], Ha Dong[†], **Xin Wang**[†], Rocco Moretti, Yu Wang, Zhaoqian Su, Jiawei Gu, Bobby Bodenheimer, Charles David Weaver, Jens Meiler, and Tyler Derr. “WelQrate: Defining the Gold Standard in Small Molecule Drug Discovery Benchmarking.” In *Advances in Neural Information Processing Systems 37, Datasets and Benchmarks Track (NeurIPS ’24)*, 53222–53236.
6. Yu Wang, Nedim Lipka, Ruiyi Zhang, Alexa Siu, Yuying Zhao, Bo Ni, **Xin Wang**, Ryan Rossi, and Tyler Derr. “Topology-aware Retrieval Augmentation for Text Generation.” In *Proceedings of the 33rd ACM International Conference on Information and Knowledge Management (CIKM ’24)*, 2442–2452.

7. Clayton W. Kosonocky, Alex M. Abel, et al. “Validation and Analysis of 12,000 AI-driven CAR-T Designs in the Bits to Binders Competitions.” *bioRxiv*, 2026. (Consortium author, Bits to Binders Competitors)

TALKS

AI Scientist Summer Workshop	Aug 2026
<i>Contributed Talk: Benchmarking AI Agents for Addressing Scientific Challenges Across Scales</i>	<i>Cambridge, MA</i>

WORK EXPERIENCE

Summer Researcher	Jun 2024 – Aug 2024
<i>Vanderbilt University, Computer Science Department</i>	<i>Nashville, TN</i>
- Analyzed ego-subgraph topology of anomalous nodes across benchmark datasets and improved graph anomaly detection by augmenting local topology with a diffusion model. - Contributed to an Amazon Research Awards proposal, <i>Generative AI for Graph Anomaly Detection</i>.	
Undergraduate Teaching Assistant	Sep 2022 – Jun 2023
<i>UC Santa Barbara, Dept. of Statistics & Applied Probability</i>	<i>Goleta, CA</i>
- Supported courses in Statistical Machine Learning, Regression Analysis, and Big Data Analytics. - Led lab sections and office hours, helping students with coursework and programming.	
Algorithm Engineer Intern	Jun 2022 – Aug 2022
<i>Ping An Technology, Intelligent City Group</i>	<i>Shenzhen, China</i>
- Mined and analyzed monthly city-level macroeconomic data and trained LSTM-based forecasting models. - Curated and preprocessed the policy database powering a policy recommendation system.	
Algorithm Intern	Jun 2021 – Aug 2021
<i>Bosera Funds, Index and Quantitative Department</i>	<i>Shenzhen, China</i>
Collected and modeled daily stock transaction data for quantitative analysis.	

TECHNICAL SKILLS

Methods: Protein & RNA Language Models, Diffusion Language Models, GNNs, LLM Fine-tuning, Agent Evaluation
Programming: Python, C/C++, SQL, R
ML Frameworks: PyTorch (DDP), PyTorch Geometric, Hugging Face Transformers, PEFT/LoRA, DeepSpeed, vLLM
AI for Biology: ESM/ESMFold, AlphaFold, RFdiffusion, ProteinMPNN, Chai-1, ViennaRNA, PyMOL
Agent Evaluation & HPC: Harbor, Terminal-Bench, Slurm, Docker, Apptainer, Git, Linux, Weights & Biases

HONORS & AWARDS

Vanderbilt Graduate School Travel Grant (\$1,000)	Nov 2024
Vanderbilt Graduate Fellowship (\$20,000)	Sep 2023
College Honors: UCSB Arts and Sciences	Jun 2023
People’s Choice Award: UCSB Math Directed Reading Program (DRP)	Jun 2022

SERVICES

Reviewer: ICLR 2026, ICLR 2027
Subreviewer: NeurIPS 2024, CIKM 2024, SDM 2024, WWW 2024, ECML-PKDD 2024
PC Member: GTA³-2024, WSDM 2024-MLoG, MLoG-GenAI (KDD 2025)
Board Member: Vanderbilt University Riichi Mahjong League
Mentor: UCSB Mentorship Program
Audit Director: Shenzhen Middle School Student Activity Association